HANDS-ON 3 · DAY 1 / AFTERNOON · GROUP'S SHEET

Grading wines: from seven grades to three lines

The case. A winery sells its wines in three lines: low, medium and high. Today each wine waits for a tasting panel before it is placed. The winery asks whether its laboratory measurements could place a new wine first, and how far such a model can be trusted, class by class.

DATA CARD · WINEQUALITY_ALL.CSV

What a row is. One sample of vinho verde, red or white, from northern Portugal, tested at the region's official certification body between May 2004 and February 2007. 6,497 rows, 13 columns, no missing values.

The target. `quality`: the median of the scores of at least three blind tasters, from 0 (very bad) to 10 (excellent). Only grades 3 to 9 occur, and 5 and 6 hold 77% of the wines.

How the rows were chosen. Wines with a protected designation of origin, sent by their producers for certification. 4,898 are white, 1,599 red. Only lab measurements are given: no grape, brand or price.

Its history. The authors modelled red and white apart, as a regression of the grade. The course file joins their two files and adds `color`. 1,177 rows (18%) are exact copies of another row; the paper does not mention them. Step 1 removes them.

Source and licence. P. Cortez, A. Cerdeira, F. Almeida, T. Matos and J. Reis (2009), Modeling wine preferences by data mining from physicochemical properties, *Decision Support Systems* 47(4):547 to 553. UCI Machine Learning Repository, dataset 186, doi:10.24432/C56S3T, CC BY 4.0.

column	what it measures	column	what it measures
`fixed acidity`	tartaric acid, g/dm³	`total sulfur dioxide`	mg/dm³, free and bound
`volatile acidity`	acetic acid, g/dm³	`density`	g/cm³
`citric acid`	g/dm³	`pH`	acidity on the pH scale
`residual sugar`	sugar left after fermentation, g/dm³	`sulphates`	potassium sulphate, g/dm³
`chlorides`	sodium chloride, g/dm³	`alcohol`	% of volume
`free sulfur dioxide`	mg/dm³	`color`	`red` or `white`
`quality`	**the target**: the grade, 0 to 10		

STEP 1 · EXPLORE THE WINES, THEN REMOVE THE COPIES

1 ☐ **Load the data**
In a new project, on the Flow: **+ Dataset** → **Upload your files**, choose `winequality_all.csv`, then **Create**.

2 ☐ **Count the grades**
Open the dataset: **Explore**, click the header of `quality`, then **Analyze**; the same for `color`.

Wines graded 3: _____ graded 9: _____ share of grade 6: _____ white wines: _____

3 ☐ **Remove the copies**
Select the dataset, then **+ Recipe** → **Distinct**. Keep every column, **Create**, then **Run**.

Rows after **Distinct**: _____ copies removed: _____

STEP 2 · TRAIN ON THE SEVEN GRADES

1 ☐ **Create the analysis**
Select the **distinct** dataset, then **Lab** → **AutoML Prediction**. Target `quality`. Keep **Quick prototypes**, then **Create**.

2 ☐ **Say that the grades are classes**
Design → **Target**: if Dataiku proposes **Regression** (the grades are numbers), set the prediction type to **Multiclass classification**.

3 ☐ **Use five folds, then train**
Design → **Train/Test set**: **K-fold cross-test**, 5 folds, stratified. Click **Train**.

Metric **Results** ranks by: _____ best model: _____ score and ±: _____

STEP 3 · READ THE MODEL CLASS BY CLASS

1 ☐ **The confusion matrix**
Results → **model** → **Performance** → **Confusion matrix**. Rows are actual grades, columns predicted ones.

Grades never predicted: _____ share of the 6s predicted 6: _____ of the 7s predicted 7: _____

2 ☐ **One ROC curve per grade**
Performance → **ROC & PR curves**: choose the class 8.

AUC of grade 8: _____ How can a grade be ranked well and yet never be predicted?

4 · THREE LINES INSTEAD OF SEVEN GRADES

1 ☐ **Regroup the grades**
On the distinct dataset: **+ Recipe** → **Prepare**, add a **Formula** step with output column `grade`, then run the recipe:

```
if(quality <= 5, "low", if(quality == 6, "medium", "high"))
```

2 ☐ **Train on the lines**
On the prepared dataset: a new **Lab** → **AutoML Prediction**, target `grade`, the settings of Step 2. **Train**.

Best score: _____ All at 1.000? **Design** → **Features handling**: which column gives `grade` away? _____

3 ☐ **Remove the leak**
Reject that column, and **Train** again.

Best score now, and its ±: _____

5 · DECIDE FOR THE WINERY

1 ☐ **Put the two models side by side**
Step 2 predicts seven grades, panel 4 three lines. For the best model of each: **Performance** → **Detailed metrics**, and **Confusion matrix** for the last row.

	Seven grades (Step 2)	**Three lines** (panel 4)
Accuracy: wines in the right class (slide 6: micro)		
F1: one per class, averaged by Dataiku		
Top wines found: % of actual 8s (highs) predicted 8 (high)		

2 ☐ **Find how Dataiku averages the F1**
Design → **Metrics**. Weighted by class size, grades 5 and 6 (three wines in four) decide the average; unweighted, grade 8 counts as much as grade 6.

☐ weighted by class size (slide 6: weighted) ☐ every class the same (slide 6: macro)

3 ☐ **Decide**
The winery wants each new wine placed in a line before the tasting panel. Which model would it deploy, and which number of the table should it watch? Why that one?

If time allows. (a) Merge only the rare grades, 3 into 4 and 9 into 8: Formula `if(quality == 3, 4, if(quality == 9, 8, quality))`, Multiclass again, `quality` rejected. Does the F1 rise?
(b) **Design** → **Train/Test set**, **Sampling method**: **Class rebalance**, train again. The rare grades' recall, and accuracy? Does the score still describe the wines the winery makes?